

GOVT 6019: Rolling Dice

Week 5 in-class activity · Fall 2026

Names: _____

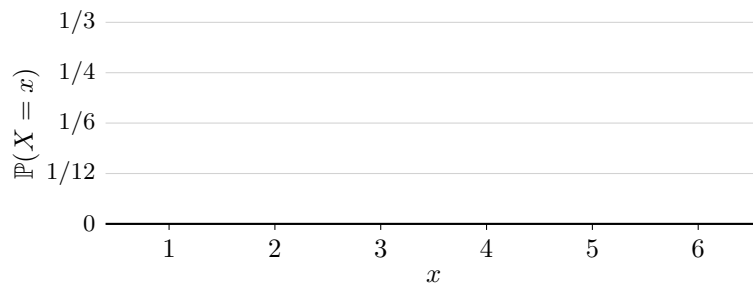
Work in pairs. You need one six-sided die, a pencil and this sheet.

Part 1. Before you roll

Let X be the result of one roll of a fair die. You worked out this distribution in Week 4.

$\mathbb{E}[X] =$ _____ $\mathbb{V}[X] =$ _____

Figure A. Draw the probability mass function of X .



Part 2. Roll

One trial:

1. Roll the die **four** times and add the four results.
2. Write the sum in the next cell of the log below.
3. Find that sum on the axis of Figure B (next page). Put an $\times$ in the lowest empty box above it.

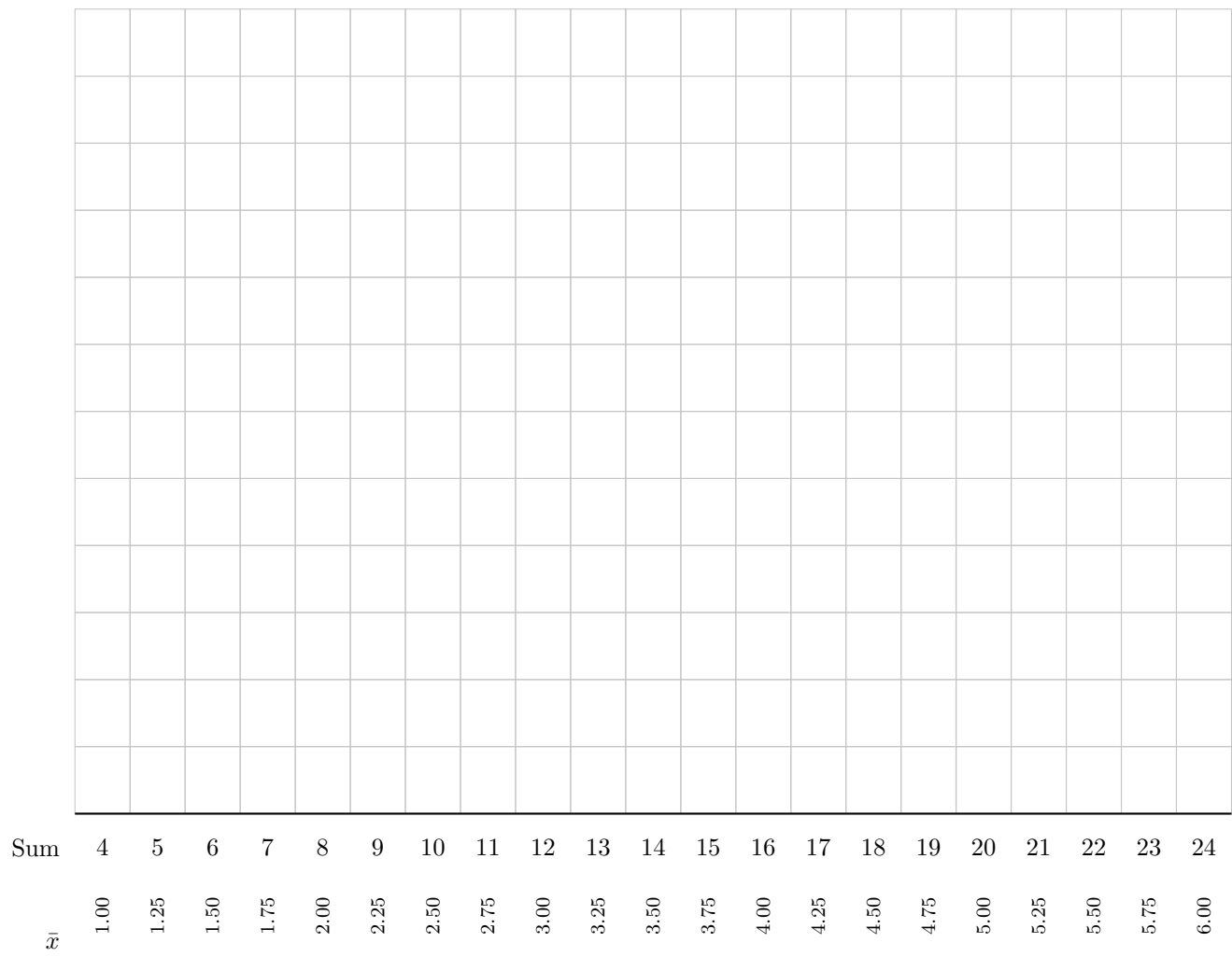
The number printed under each sum is $\bar{x}$, the mean of the four rolls: the sum divided by 4.

Do **40 trials**.

Log of sums

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40

Figure B. The means of your 40 trials. Each × is one trial.



Part 3. Questions

Recall the premise of the Central Limit Theorem (CLT): the sampling distribution of the sample means (i.e., the distribution of $\bar{x}$'s) will follow a Normal distribution with mean μ and standard deviation $\sigma/\sqrt{n}$, regardless of the distribution of the population. Because we know the distribution of the values of the die and this distribution is symmetric, we can see how the CLT works with a small sample size ($n = 4$).

We will now compare the results of rolling one die versus the activity you performed (the mean of four rolls of a die).

1. Comment on the difference in shape between Figure A and Figure B.
2. Do you believe that the Central Limit Theorem is working with regards to the shape of Figure B?
3. If the population has a standard deviation of σ and the sampling distribution of the sample means has a standard deviation of $\sigma/\sqrt{n}$ (according to the CLT), which one of these two values will be lower if $n > 1$? In other words, which of these two distributions do you expect to have lower variability: the population distribution or the sampling distribution of the sample means?